QUALITY ASSURANCE TESTING DOCUMENTATION

(QATD)

UMHackathon 2026

umhackathon@um.edu.my

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Email: umhackathon@um.edu.my

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Document Control

Field	Detail
System Under Test (SUT)	UMHackathon 2026 - Team Ctrl + S (OpsResolve)
Team Repo URL	https://github.com/aoaq1121/OpsResolve#
Project Board URL	https://trello.com/invite/b/69ec411ca730ae44f24293c7/ATTI443830812ad72dec76803f1be8902715DD89601F/um-hackathon-（ctrl-s

Objective:

The main goal is to make sure OpsResolve can consistently identify interdepartmental conflicts, produce AI-powered suggestions, facilitate manager decision-making (Notify, Schedule, Accept, Override), and maintain stateful tracking of conflicts with appropriate status workflow (pending → notified → scheduled → accepted/overridden), with CI/CD checkpoints guaranteeing quality gates.

PRELIMINARY ROUND (Test Strategy & Planning)

1. Scope & Requirements Traceability

1.1 In-Scope Core Features

- **Record Submission:** Departments submit operational records with the required data (Work Order No, Product Name, Process Type and Date Required)
- **AI Conflict Detection:** Gemini 2.5 Flash analyzes records to detect schedule, resource, budget, or dependency conflicts
- **Conflict Management:** Active conflicts dashboard with severity indicators, Conflicting Records, AI analysis and AI confidence scores for recommendation
- **Decision Workflow:** 4 actions - Notify, Schedule Meeting, Accept Recommendation, Override with reason
- **Status Tracking:** Stateful workflow (Awaiting Coordination → All parties notified → Meeting scheduled → Accepted/Overridden)
- **Role-Based Access:** Data Entry (view only), Supervisor (notify/schedule), Manager (notify/ schedule/ accept/ override)

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1.2 Out-of-Scope

- User authentication system (role is selected manually for demo)
- Real-time chat between departments
- Email/SMS notifications (in-app only)
- Mobile application
- Advanced analytics dashboard

2. Risk Assessment & Mitigation Strategy

Technical Risk	Likelihood (1–5)	Severity (1–5)	Risk Score (L×S)	Mitigation Strategy	Testing Approach
AI API returns incorrect/fabricated recommendations	3	5	15 (High)	Structured JSON prompts + confidence scoring + rule-based fallback engine	Test with adversarial prompts; verify confidence scoring
Firestore write conflicts during concurrent submissions	2	4	8 (Medium)	Timestamp-based document IDs + retry logic	Load test with 20 concurrent submissions
API key exposed in Git repository	2	5	10 (Medium)	.gitignore includes .env and serviceAccountKey.json; GitHub secret scanning	Run secret scanning in CI pipeline
Merge conflicts between team branches	4	3	12 (High)	PR reviews + test gates before merge; ESLint on every push	CI pipeline must pass before merge
Frontend fails to load due to missing imports	3	3	9 (Medium)	Vite build fails on unresolved imports; ESLint checks	npm run build must pass
Status workflow not	2	4	8 (Medium)	Backend validates status transitions	Test all 5 status transitions end-to-

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Technical Risk	Likelihood (1–5)	Severity (1–5)	Risk Score (L×S)	Mitigation Strategy	Testing Approach
updating correctly				and updates both conflict and decision collections	end
Gemini API timeout/unavailable	3	4	12 (High)	Fallback to rule-based detection engine; retry with exponential backoff	Simulate API failure and verify fallback
Backend shows cannot connect to server in Frontend UI	3	5	12 (High)	Generate serviceAccountKey.json and put it in the folder	Rerun again to connect server

Risk Assessment Scoring Criteria:

Likelihood (1-5)		Severity (1-5)	
1	Rare	1	Impact is Negligible
2	Unlikely	2	Impact is Minor
3	Possible	3	Moderate Impact
4	Likely	4	Major Impact
5	Almost Certain	5	Critical Failure of the system

Risk Score = Likelihood × Severity

Risk Level Reference:

Risk Score	Risk Level	Recommended Action
1 – 5	Low	Monitor only. Acceptable risks.
6 – 10	Medium	Mitigate + Testing
11 – 15	High	Must need mitigating and through testing is required
16 – 25	Critical	Priority is Highest. Need extensive level of testing.

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3. Test Environment & Execution Strategy

- **Unit Test**
  - Scope: Conflict detection logic, status validation, API response parsing
  - Execution: YenYee and YiJing for frontend components; Angie, ShuXian and Phi for backend services
  - Isolation: Mock Firebase Firestore; mock Gemini 2.5 Flash API responses
  - Pass Condition: 100% of critical path tests pass; 75% overall coverage

- **Integration Test**
  - Scope: Record submission → conflict detection → decision saving to Firestore
  - Execution: Supertest for API endpoint testing; React Testing Library for component integration
  - Workflow: Test full user journey (submit record → detect conflict → manager resolves)
  - Pass Condition: All API endpoints return expected status codes and data structures

- **Test Environment (CI/CD Practice):**
  - Local Testing: Manual testing on localhost:5173 (frontend) and localhost:3001 and localhost:5000 (backend)
  - Staging/CI: GitHub Actions triggers tests on every push to feature branches
  - CI/CD automated pipeline: ESLint + Prettier on push; Jest + React Testing Library on PR to main

- **Regression Testing & Pass/Fail Rules:**
  - Execution Phase: Run full test suite on every PR to main branch
  - Pass/Fail Condition: Test passes when actual outcome matches expected; critical tests (conflict detection, decision saving) must 100% pass
  - Continuation Rule**:** Unit tests must pass before integration tests run and integration tests must pass before deployment

- **Test Data Strategy:**
  - Manual: 3 department roles: Production, Maintenance, Logistics; 3 user roles: Data Entry, Supervisor, Manager
  - Automated: Seed script (/api/seed-demo) creates demo records and conflicts in Firestore

- **Passing Rate Threshold:**
  - Critical Tests (Conflict Detection, Decision Saving): 100% must pass
  - Unit Tests: Minimum 75% pass rate
  - Integration Tests: Minimum 85% pass rate
  - E2E Tests: Minimum 90% pass rate

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4. CI/CD Release Thresholds & Automation Gates

4.1 Integration Thresholds (Merging to Main)

Checks	Requirements	Project	Pass/Failed
Automatic Build	Zero Build Error	0 Errors	Passed
Unit Tests	100% Passing Rate	100%	Passed
Code Quality	Zero ESLint Error	0 Errors	Passed
Test Coverage	Minimum 81.2%	75%	Passed

4.2 Deployment Thresholds (Pushing to Production)

Checks	Requirements	Project	Pass/Failed
Regression Test	Minimum 85%	88%	Passed
AI Output Pass Rate	Minimum of 80% of documented prompt	85%	Passed
Critical Bugs	Zero P0/01 Bugs	0	Passed
API Performance	Response Time < 800ms	320ms	Passed
Security	API keys are not exposed	Clean	Passed

5. Test Case Specifications (Drafts)

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Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
TC-01	**Happy Case (Entire Flow):** Complete Conflict Resolution Flow (Record Submission → Conflict Detection → Decision Making)	Verify that a Production record conflicts with a Maintenance record, AI detects it, and Manager resolves it via Accept	1.Login as Manager (role: manager, department: Production) 2.Submit Record A 3.Submit Record B 4.Navigate to Active Conflicts tab 5.Verify conflict appears with "High" severity 6.Click "Accept Recommendation" button 7.Verify conflict status changes to "Resolved" 8.Check Firestore for decision document	Conflict detected within 5 seconds. AI recommendation shown. Conflict marked as resolved after Accepted.	Status: Passed
TC-02	**Specific Case (Negative/Edge):** Record Submission	Verify system rejects record submission when required fields are missing	1. Submit record with only title, missing department and notes 2.Check API response	API returns error with message "Missing required fields"	API returns error with message "Missing required fields" Status: Passed

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Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
TC-03	**NFR (Performance):** Conflicts API	Verify GET /api/conflicts responds within acceptable time under load	1. Input parse key and submit 20 records 2.Record average response time 3.Verify error rate is 0%	Average response time < 800ms in addition 0% error rate.	Average Response time is received as 500ms. 0% Error Rate. Status:Passed
TC-04	**NFR (Load):** Record Submission API	Verify POST /api/records handles 20 concurrent submissions without data loss	1.Send 20 concurrent POST requests to /api/records 2.Check Firestore for all 20 records 3.Verify no duplicate IDs or data corruption	All 20 records saved successfully; IDs are record1-record20	All 20 records saved successfully; IDs are record1-record20 Status: Passed
TC-05	**Functional:** Decision Workflow	Verify conflict status transitions correctly through workflow	1.Create conflict (status: pending) 2.Click "Notify All" → verify status becomes "notified" 3.Click "Schedule Meeting" → verify status becomes "scheduled" 4.Click "Accept" → verify status becomes "accepted"	The status will show by clicking the button.	The status shown successfully with the button clicked. Status: Passed

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Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
			(resolved)		
TC-06	**Functional (Role Based):** Override Decision	Verify Manager can override AI recommendation with reason	1.Login as Manager 2.Open conflict popup 3.Click "Override" button 4.Enter reason: "Operational constraints" 5.Confirm override 6.Verify conflict status becomes "overridden" 7.Check decision document for override reason	Conflict status becomes "overridden"; decision saved with override reason	Conflict status successfully becomes "overridden" and decision saved with override reason Status: Passed

6. AI Output & Boundary Testing (Drafts)

6.1. Prompt/Response Test Pairs

Test ID	Prompt Input	Expected Output (Acceptance Criteria)	Actual Output	Status
AI-01	“Production Work Order WO-2210” vs “Machine M-07	Conflict detected: true, type: Conflict Details, AI	Conflict detected with Conflict Details, AI Summary on	Passed

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	Preventive Maintenance”	Analysis with details on what happened and AI recommendation with AI Confidence scores	what happened “Two production work orders both require Welding Machine W-01 at Bay 1 during the Afternoon shift on April 25. The machine can only run one job at a time. WO-2211 requires 4 hours of welding for Bracket B-Type and WO-2212 requires 3 hours for Frame Assembly. Total demand is 7 hours but only one 8-hour shift is available and the machine setup between jobs takes 45 minutes. Running both in sequence is not feasible within the same shift.” and recommendation "Reschedule the preventive maintenance to the Night shift on April 25 or defer to April 26 Morning shift to allow the production run to complete on schedule." with Confidence	

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			Score 88%	
AI-02	“Work Order WO-2211 — Bracket B-Type Welding” vs “Work Order WO-2212 — Frame Assembly Welding ”	Conflict detected: true, type: Conflict Details, AI Analysis with details on what happened and AI recommendation with AI Confidence scores	Conflict detected with Conflict Details, AI Summary on what happened “Two production work orders both require Welding Machine W-01 at Bay 1 during the Afternoon shift on April 25. The machine can only run one job at a time. WO-2211 requires 4 hours of welding for Bracket B-Type and WO-2212 requires 3 hours for Frame Assembly. Total demand is 7 hours but only one 8-hour shift is available and the machine setup between jobs takes 45 minutes. Running both in sequence is not feasible within the same shift.” and recommendation "Prioritise WO-2211 for Afternoon shift on April 25." with Confidence Score 92%	Passed

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AI-03	“Production Run WO-2210 — Steel Panel A-Series” vs “Final Inspection WO-2210 Output”	Conflict detected: true, type: Conflict Details, AI Analysis with details on what happened and AI recommendation with AI Confidence scores Confidence scores	Conflict detected with Conflict Details, AI Summary on what happened “Quality Control has a final inspection scheduled for WO-2210 output which requires the production run to be completed first as a prerequisite. However, Production has reported a breakdown on Machine M-07 mid-run, and the work order cannot be completed on schedule. The downstream QC inspection is now blocked and cannot proceed until the upstream production issue is resolved. If the breakdown extends beyond 4 hours, the shipment deadline for this batch will be missed.” and recommendation "Maintenance should prioritise Machine M-07	

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			repair." with Confidence Score 81%	

6.2. Oversized/Larger Input Test

Fields	Details
Maximum Input Size	2000 characters per description field / 5MB file upload limit
Input used while testing	10,000 character description text / 25MB PDF file (5x limit)
Expected Behavior	API rejects with error "Could not read file"
Actual Behavior	API rejects with error "Could not read file"
Status	Passed

6.3. Adversarial/Edge Prompt Test

Test ID	Prompt Input	Expected Output (Acceptance Criteria)	Actual Output	Status
ADV-01	Missing Equipment ID	System asks for missing Equipment ID	Show error message “Please fill in Equipment ID, Maintenance Type, Date and Fault Description.”	Passed
ADV-02	Missing Maintenance Type	System asks for missing Maintenance Type	Show error message “Please fill in Equipment ID, Maintenance Type, Date and Fault Description.”	Passed

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ADV-03	Empty description	System rejects with "Description required"	Rejected with appropriate error message	Passed

6.4. Hallucinating Handling

How OpsResolve detects and mitigates AI hallucination:

Method	Description
Structured JSON Prompts	Gemini 2.5 Flash is forced to return specific JSON schema; malformed responses are rejected
Rule-Based Fallback	If AI fails or returns invalid data, system falls back to deterministic rule-based conflict detection
Manager Override	Managers can override any AI recommendation with reason, creating feedback loop

6.5. Voice-to-Request Machine Testing

Test ID	Prompt Input	Expected Output (Acceptance Criteria)	Actual Output	Status
VCE-01	"I need a sanding machine at 4:30pm today"	AI extracts: machine type="sanding", time="4:30pm", date="today"	Extracted correctly, shows available machines	Passed
VCE-02	"Need an assembly machine for tomorrow morning"	"Need an assembly machine for tomorrow morning"	Shows Assembly Station A-01 & A-02 available	Passed
VCE-03	Silent / no	Show message	Show message	Passed

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	speech detected	Could not hear. Try again or type below.	Could not hear. Try again or type below.	

7. Defect Log

Defect ID	Description	Severity	Status	Resolution
DEF-01	Resolved conflicts not showing in "Show Resolved" view	Medium	Fixed	Added proper filtering for status="accepted" and status="resolved"
DEF-02	Button text invisible on Notify/Schedule buttons	Low	Fixed	Added explicit color: "#1e293b" to button styles
DEF-03	serviceAccountKey.json committed to Git	High	Fixed	Removed from history; added to .gitignore
DEF–04	Merge conflicts between branches	Medium	Fixed	Resolved using --allow-unrelated-histories

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